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Late Stage Functionalization of Secondary Amines via a Cobalt-Catalyzed Electrophilic Amination of Organozinc Reagents

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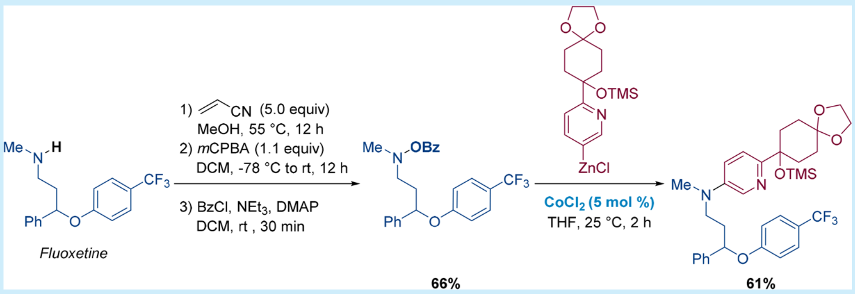
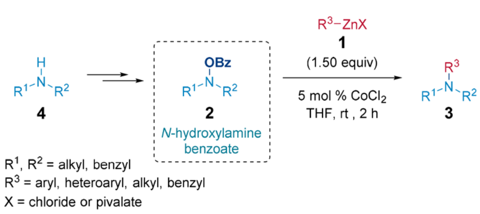
Supporting Information



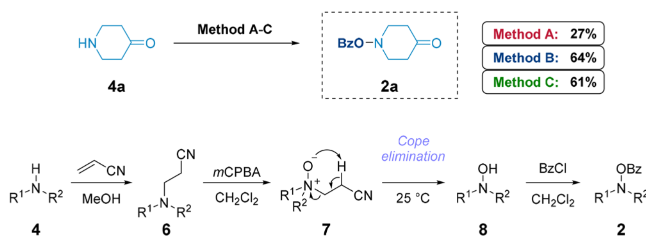
ABSTRACT: A general preparation of polyfunctional hydroxylamine benzoates from the corresponding secondary amines is reported. This convenient synthesis allows the setup of a late-stage functionalization of various secondary amines, including pharmaceuticals and peptidic derivatives. Thus, a cross-coupling of hydroxylamine benzoates with various alkyl-, aryl-, and heteroaryl-zinc chlorides in the presence of 5 mol % $CoCl_2$ (25 °C, 2 h) provides a range of polyfunctional tertiary amines. This method was used to prepare penfluridol and gepirone.

The formation of a carbon−nitrogen bond is one of the most important reactions for the elaboration of pharmaceuticals and agrochemicals.[1] Especially the late-stage functionalization[2] of secondary amines would be a valuable method for producing new biologically active compounds.[3] Palladium-catalyzed nucleophilic aminations have tremendously improved the performance of aryl and heteroaryl aminations.[4] However, electrophilic aminations pioneered by Johnson[5] are a valuable alternative because cheaper and less toxic metal catalysts of Cu, Ru, Ni, Fe, and Co may be used.[6] Recently, we reported a new cobalt-catalyzed electrophilic amination of organozinc pivalates[7] of type **1** with *N*-hydroxylamine benzoates **2**, allowing the preparation of various functionalized amines of type **3** (Scheme 1).[8] The required *N*-hydroxylamine benzoates **2** have been prepared from the corresponding amines **4** using benzoylperoxide (BPO), as previously reported.[5] However, the scope of preparation of such *N*-hydroxylamine benzoates **2** is quite limited and

Scheme 1. Cobalt-Catalyzed Electrophilic Amination Using Organozinc Reagents 1 and *N*-Hydroxylamine Benzoates 2 Leading to Functionalized Tertiary Amines of Type 3



Scheme 2. Comparison of Three Preparations of *N*-Hydroxylamine Benzoate 2a from Amine 4a[11a]



[a]Method A: amine **4** (1.5 equiv), BPO (1.0 equiv), K_2HPO_4 (1.5 equiv), DMF, rt, 12 h. Method B: (1) amine **4** (1.0 equiv), DMDO (1.05 equiv), acetone, 0 °C, 1 h; (2) BzCl (1.2 equiv), NEt_3 (1.5 equiv), DMAP (1 mol %), CH_2Cl_2, 0 °C, 30 min. Method C: (1) amine **4**, (1.0 equiv), acrylonitrile (5.0 equiv), MeOH, 55 °C, 12 h; (2) *m*CPBA (1.1 equiv), CH_2Cl_2, −78 °C to rt, 12 h; (3) BzCl (1.2 equiv), NEt_3 (1.5 equiv), DMAP (1 mol %), CH_2Cl_2, 0 °C, 30 min.

considerably reduces the synthetic potential of these electrophilic aminations.

Herein, we report a new method with a broad scope to prepare *N*-hydroxylamine benzoates **2** and demonstrate their utility for the performance of late-stage functionalizations of various amines, including pharmaceuticals and peptides. Furthermore, we show the efficiency of this method for the preparation of the two drugs gepirone[9] (**3p**) and penfluridol[10] (**5**).

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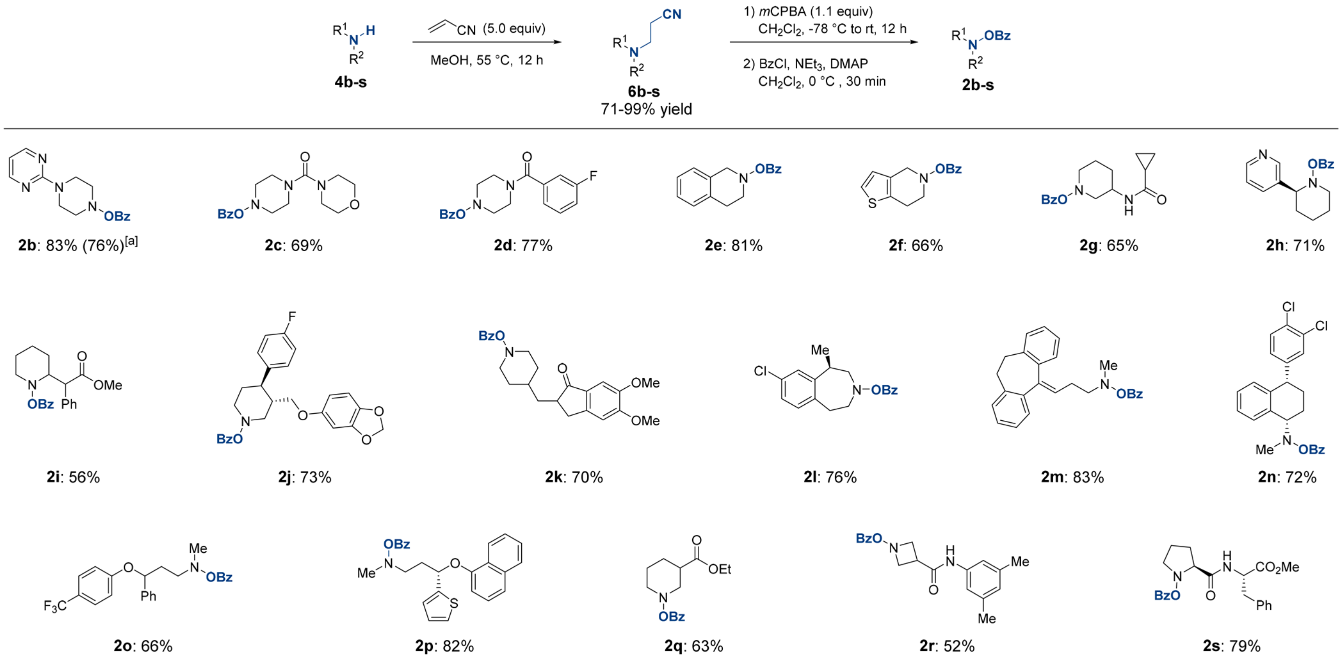
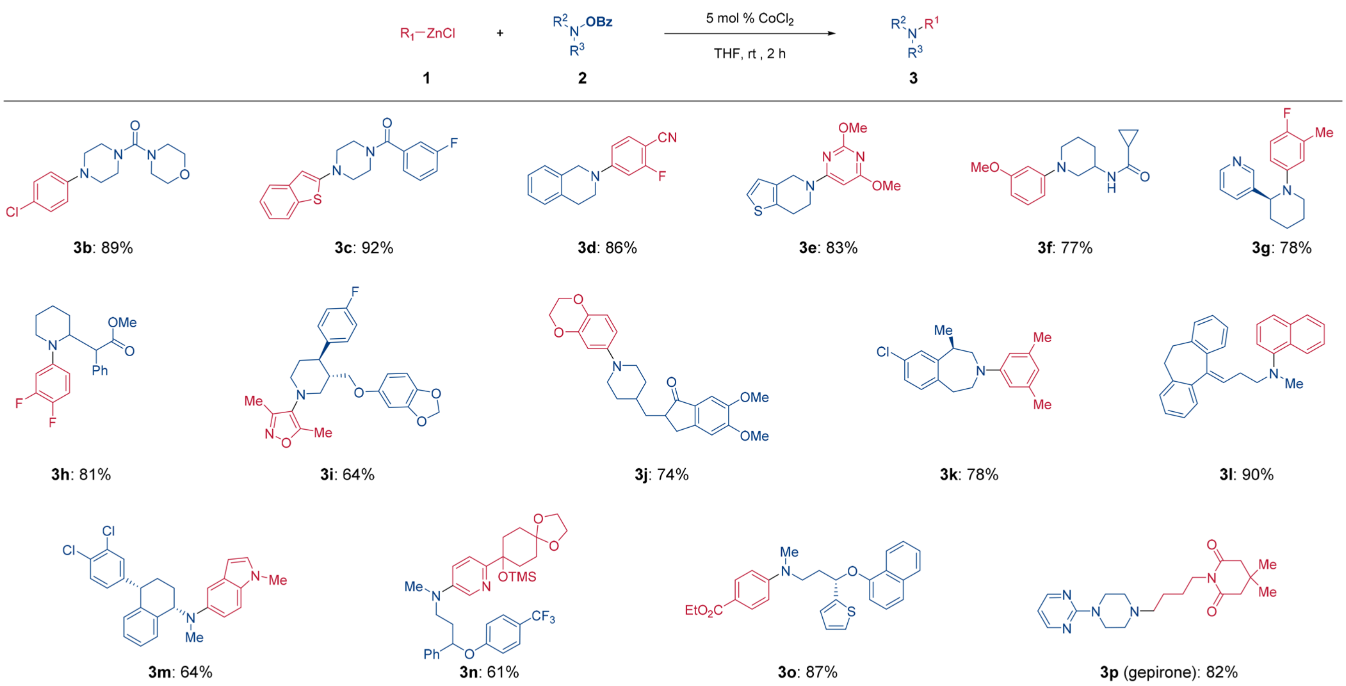
Scheme 3. Preparation of *N*-Hydroxylamine Benzoates[a]



[a]All experiments were performed on a 1 mmol scale and the reported yields are isolated yields of analytically pure products; [a] 10 mmol scale.

Scheme 4. Electrophilic Amination of Organic Chlorides of Type 1[a]



[a]All experiments were performed on a 1 mmol scale, and the reported yields are isolated yields of analytically pure products.

Preliminary experiments have shown that the benzoyloxylation of a typical functionalized amine such as 1,4-piperidone (**4a**) provides the corresponding benzoyloxyamine **2a** in only 27% yield using BPO (Method A, Scheme 2). An alternative method using a dimethyldioxirane (DMDO)[12] oxidation and subsequent benzoylation with PhCOCl (BzCl) provides **2a** in 64% yield (Method B). Unfortunately, this reaction could not be easily scaled up due to low yields for the preparation of DMDO. Several other oxidation methods were tested, but were neither selective nor high yielding. However, the oxidation method of O'Neil for preparing *N*-hydroxylamines **8** proved to be convenient and general.[13] According to this method, amine **4** was treated with acrylonitrile (MeOH, 55 °C, 12 h), providing a tertiary amine of type **6**. Its oxidation with *meta*-chloroperbenzoic acid (*m*CPBA; CH_2Cl_2, −78 °C, 3 h) gives an amine *N*-oxide of type **7**, which underwent a Cope

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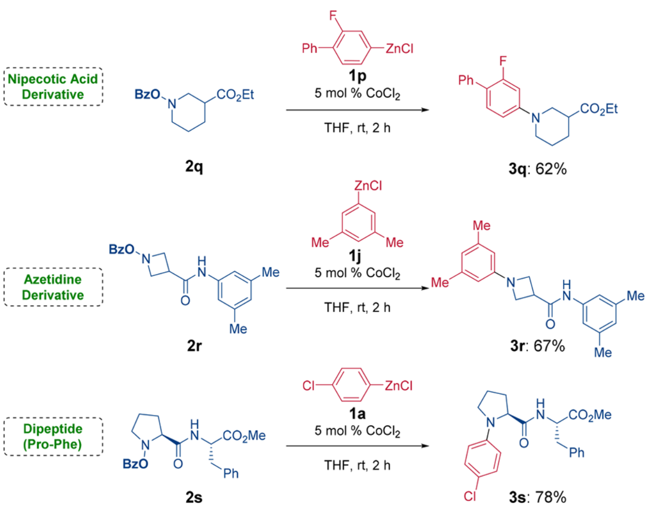
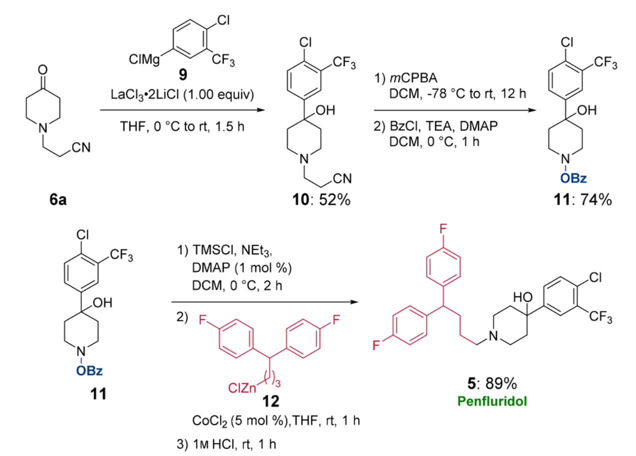
Scheme 5. Functionalization of Amino Acids 2q and 2r and Peptidic Substrates 2s Using the Cobalt-Catalyzed Amination of Organozincs 1c, 1q, and 1r



Scheme 6. A New Synthesis of Penfluridol 5



elimination (25 °C, 9 h), affording *N*-hydroxylamines of type **8**.[13] Benzoylation with BzCl (NEt_3, DMAP, CH_2Cl_2, 0 °C, 0.5 h) furnished the desired hydroxylamine benzoates **2**. Applying this sequence to **4a** led to the desired product **2a** in an overall yield of 61% (Method C).

This selective oxidation was performed on diverse amines **4b–s** affording via the 2-cyanoethyl amines **6b–s** the desired hydroxylamine benzoates in satisfactory overall yields (Scheme 3). In contrast to Method B, Method C allows a convenient scale-up. Thus, the preparation of the hydroxylamine benzoate **2b** derived from a piperazine (83% yield on 1 mmol scale) could readily be scaled-up (77% yield on 10 mmol scale; Scheme 3). Likewise, we prepared the related piperazine and piperidine derived hydroxylamine benzoates **2c–2g** (65–81% yields). This method has been extended to biologically active amines, such as the alkaloid anabasine and the psychostimulant methylphenidate and led to the expected hydroxylamine benzoates **2h** and **2i** in 56–71% yields. Important drugs bearing cyclic secondary amines such as paroxetine (anti-depressant), debenzylated donepezil (treatment of dementia), and lorcaserine (former morbid obesity medication) were smoothly converted to the desired products **2j–2l** in 73–76% yields. Also, open-chain antidepressants like nortriptyline, sertraline, fluoxetine, and duloxetine were predictably converted to the hydroxylamine esters **2m–2p** in 66–90% yields. Finally, amino acids and a peptide such as a nipecotic acid derivative, an azetidine derivate, and a dipeptide were chemoselectively converted to the corresponding hydroxylamines **2q–2s** in 52–79% yields.

Having in hand a general conversion of secondary amines **4** to the corresponding hydroxylamine benzoates **2**, we have shown their cross-coupling with polyfunctional organozinc chlorides[14] provides a wide range of tertiary amines of type **3**. Thus, various aryl- and heteroaryl-zinc chlorides **1a–1d** underwent a cobalt-catalyzed electrophilic amination with cyclic hydroxylamine benzoates **2c–2f** in the presence of 5 mol % $CoCl_2$ (THF, 25 °C, 2 h), leading to polyfunctional tertiary amines **3b–3e** in 83–92% yields (Scheme 4). This amination was found to be compatible with several important functional groups (secondary and tertiary alcohols, primary and secondary amines, amides and epoxides).[15,16] For example, the reaction of 3-anisylzinc chloride (**1e**) with hydroxylamine ester **2g**, bearing an acidic amide proton proceeded smoothly in the presence of 5 mol % $CoCl_2$, affording the tertiary amine **3f**. These electrophilic aminations complement the nucleophilic amination of Buchwald and Hartwig.[4] Thus, the melatonin receptor ligand **3f** as well as the 517-β-hydroxysteroid dehydrogenase inhibitor **3b** have been prepared via the nucleophilic Pd-catalyzed amination in moderate yields (respectively 55[17] and 37%[18]), whereas the present electrophilic amination produces these pharmaceutical targets in 89 and 77% yield. The robustness of this amination has been used for completing a late-stage functionalization of several alkaloids and pharmaceuticals bearing a secondary amine. Thus, the corresponding hydroxylamine benzoates **2h–2p** provided, after treatment with the organozinc chlorides **1f–1n**, the desired functionalized pharmaceuticals or alkaloids **3g–3o** in 61–90% yields. Alkylzinc chlorides such as **1o**[19,20] are also good reaction partners for this amination. Hence, zinc reagent **1o** was aminated with **2b**, providing gepirone (**3p**), an antidepressant drug, in 82% yield.

Encouraged by these results, we envisioned a late stage functionalization of amino acids and peptides. Therefore, the hydroxylamine benzoates derived from two β-amino-acids **2q** and **2r** and dipeptide **2s** underwent the expected amination with organozinc chlorides **1p**, **1j**, and **1a**, leading to the arylated amino-acids **3q–3r** and peptide **3s** in 62–78% yield (Scheme 5). As observed for benzoate **2g**, the amidic proton of **2r** and **2s** did not disturb the desired amination.

To demonstrate the synthetic versatility of this amination, we also performed a short synthesis of penfluridol (**5**), a highly potent antipsychotic (Scheme 4). Thus, the arylmagnesium chloride **9** underwent a $LaCl_3 \cdot 2LiCl$[21] mediated addition to the 2-cyanoethyl-piperidone (**6a**)[19] (THF, 25 °C, 1.5 h), leading to the tertiary alcohol **10** in 52% yield. Without protection of the tertiary hydroxyl function, alcohol **10** was converted by the standard method to the hydroxylamine benzoate **11** in 74% yield. Protection of **10** with TMSCl followed by the cobalt-catalyzed amination using the alkylzinc chloride **12**[19] and desilylation (1M HCl, 25 °C, 1 h) produces penfluridol (**5**) in 89% yield (Scheme 6).

In summary, we reported a very general and functional group tolerating synthesis of hydroxylamine benzoates **2** and demonstrated their utility in the cobalt-catalyzed amination of various alkyl-, aryl-, and heteroaryl-zinc chlorides, leading for the first time to complex polyfunctional amines in a predictable

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way. We showed that this method allows the late stage functionalization of various drugs, including peptidic target substrates, and can be readily used for the synthesis of complex target amines such as the pharmaceuticals penfluridol (**5**) and gepirone (**3p**). Further extensions are currently studied in our laboratories.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.8b03787.

General experimental procedures, detailed synthetic procedures, and analytical data for all compounds mentioned (PDF)

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Notes

The authors declare no competing financial interest.

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